STA2002 Tutorial 10

1 Summary

1.1 One-way ANOVA

Suppose $X_{ij} \overset{i.i.d.}{\sim} N(\mu_i, \sigma^2)$, $j = 1, 2, \ldots, n_i$, $i = 1, 2, \ldots, m$, $n = \sum_{i=1}^{m} n_i$. We would like to test

$$H_0 : \mu_1 = \mu_2 = \ldots = \mu_m$$

The test process:

$$SS(T) = \sum_{i=1}^{m} n_i(\bar{X}_{i\cdot} - \bar{X}_{\cdot\cdot})^2$$

$$SS(TO) = \sum_{i=1}^{m}\sum_{j=1}^{n_i}(X_{ij} - \bar{X}_{\cdot\cdot})^2$$

$$SS(E) = SS(TO) - SS(T) = \sum_{i=1}^{m}\sum_{j=1}^{n_i}(X_{ij} - \bar{X}_{i\cdot})^2$$

Under H_0, see ANOVA table:

Table 9.3-2 Analysis-of-variance table

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	F Ratio
Treatment	SS(T)	$m-1$	$\text{MS(T)} = \frac{\text{SS(T)}}{m-1}$	$\frac{\text{MS(T)}}{\text{MS(E)}}$
Error	SS(E)	$n-m$	$\text{MS(E)} = \frac{\text{SS(E)}}{n-m}$	
Total	SS(TO)	$n-1$		

1.2 Two-way ANOVA, without interaction, $c = 1$

Suppose $X_{ij} \overset{i.i.d.}{\sim} N(\mu_{ij}, \sigma^2)$, $\mu_{ij} = \mu + \alpha_i + \beta_j$, $i = 1, 2, \ldots, a$, $j = 1, 2, \ldots, b$. We would like to test

$$H_0^A : \alpha_i = 0 \text{ for } i = 1, 2, \ldots, a \quad H_1^A : H_0^A \text{ is not true}$$

$$H_0^B : \beta_j = 0 \text{ for } j = 1, 2, \ldots, b \quad H_1^B : H_0^B \text{ is not true}$$

The test process:

$$SS(A) = b\sum_{i=1}^{a}(\bar{X}_{i\cdot} - \bar{X}_{\cdot\cdot})^2$$

$$SS(B) = a\sum_{j=1}^{b}(\bar{X}_{\cdot j} - \bar{X}_{\cdot\cdot})^2$$

$$SS(TO) = \sum_{i=1}^{a}\sum_{j=1}^{b}(X_{ij} - \bar{X}_{\cdot\cdot})^2$$

$$SS(E) = SS(TO) - SS(A) - SS(B)$$

Under H_0^A and H_0^B, see ANOVA table:

Table 9.4-1 Two-way ANOVA table, one observation per cell

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	F
Factor A (row)	SS(A)	$a-1$	$\text{MS(A)} = \frac{\text{SS(A)}}{a-1}$	$\frac{\text{MS(A)}}{\text{MS(E)}}$
Factor B (column)	SS(B)	$b-1$	$\text{MS(B)} = \frac{\text{SS(B)}}{b-1}$	$\frac{\text{MS(B)}}{\text{MS(E)}}$
Error	SS(E)	$(a-1)(b-1)$	$\text{MS(E)} = \frac{\text{SS(E)}}{(a-1)(b-1)}$	
Total	SS(TO)	$ab-1$		

1.3 Two-way ANOVA, with interaction, $c > 1$

Suppose $X_{ijk} \overset{i.i.d.}{\sim} N(\mu_{ij}, \sigma^2)$, $\mu_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij}$, $i = 1, 2, \ldots, a, j = 1, 2, \ldots, b$, $k = 1, \ldots, c$ (c repeats in each cell). We would like to test

$$H_0^A : \alpha_i = 0 \text{ for } i = 1, 2, \ldots, a \quad H_1^A : H_0^A \text{ is not true}$$

$$H_0^B : \beta_i = 0 \text{ for } j = 1, 2, \ldots, b \quad H_1^B : H_0^B \text{ is not true}$$

$$H_0^{AB} : \gamma_{ij} = 0 \text{ for } i = 1, 2, \ldots, a, j = 1, \ldots, b \quad H_1^{AB} : H_0^{AB} \text{ is not true}$$

The test process:

$$SS(A) = bc \sum_{i=1}^{a} (\bar{X}_{i\cdot\cdot} - \bar{X}_{\cdots})^2$$

$$SS(B) = ac \sum_{j=1}^{b} (\bar{X}_{\cdot j\cdot} - \bar{X}_{\cdots})^2$$

$$SS(TO) = \sum_{i=1}^{a} \sum_{j=1}^{b} \sum_{k=1}^{c} (X_{ijk} - \bar{X}_{\cdots})^2$$

$$SS(E) = \sum_{i=1}^{a} \sum_{j=1}^{b} \sum_{k=1}^{c} (X_{ijk} - \bar{X}_{ij\cdot})^2$$

$$SS(AB) = SS(TO) - SS(A) - SS(B) - SS(E)$$

Under H_0^A, H_0^B and H_0^{AB}, see ANOVA table:

Table 9.4-4 Two-way ANOVA table, c observations per cell

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	F
Factor A (row)	SS(A)	$a-1$	$\text{MS(A)} = \frac{\text{SS(A)}}{a-1}$	$\frac{\text{MS(A)}}{\text{MS(E)}}$
Factor B (column)	SS(B)	$b-1$	$\text{MS(B)} = \frac{\text{SS(B)}}{b-1}$	$\frac{\text{MS(B)}}{\text{MS(E)}}$
Factor AB (interaction)	SS(AB)	$(a-1)(b-1)$	$\text{MS(AB)} = \frac{\text{SS(AB)}}{(a-1)(b-1)}$	$\frac{\text{MS(AB)}}{\text{MS(E)}}$
Error	SS(E)	$ab(c-1)$	$\text{MS(E)} = \frac{\text{SS(E)}}{ab(c-1)}$	
Total	SS(TO)	$abc-1$		

1. Let X_1, X_2, X_3, X_4 equal the cholesterol level of a woman under the age of 50, a man under 50, a woman 50 or older,and a man 50 or older, respectively. Assume that the distribution of X_i is $N(\mu_i, \sigma^2), i = 1, 2, 3, 4$. We shall test the null hypothesis $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$, using seven observations of each X_i.

x1 :	221	213	202	183	185	197	162
x2 :	271	192	189	209	227	236	142
x3 :	262	193	224	201	161	178	265
x4 :	192	253	248	278	232	267	289

Construct an ANOVA table and state your conclusion.

Source	SS	DF	MS	F	p-value
Treatment	12,280.8	3	4,093.6	3.46	0.0323
Error	28,434.5	24	1,184.77		
Total	40,715.4	27			

$F = 3.455 > F_{0.05}(3, 24) = 3.01$, reject H_0

2. A psychology student was interested in testing how food consumption by rats would be affected by a particular drug. She used two levels of one attribute, namely, drug and placebo, and four levels of a second attribute,namely, male (M), castrated (C), female (F), and ovariectomized (O). For each cell, she observed five rats. The amount of food consumed in grams per 24 hours is listed in the following table:

	M	C	F	O
Drug	22.56	16.54	18.58	18.20
	25.02	24.64	15.44	14.56
	23.66	24.62	16.12	15.54
	17.22	19.06	16.88	16.82
	22.58	20.12	17.58	14.56
Placebo	25.64	22.50	17.82	19.74
	28.84	24.48	15.76	17.48
	26.00	25.52	12.96	16.46
	26.02	24.76	15.00	16.44
	23.24	20.62	19.54	15.70

(a) Fullfill the following ANOVA table

Source	SS	df	MS	F
FactorA	29.3			
FactorB	454.7			
FactorA & B	27.3			
Error	169.3			
Total				

(b) Use the 5% significance level and test H_{AB}: $\gamma_{ij} = 0$, $i = 1, 2$, $j = 1, 2, 3, 4$.

(c) Use the 5% significance level and test $H_A : \alpha_1 = \alpha_2 = 0$.

(d) Use the 5% significance level and test $H_B : \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0$.

(a) The ANOVA table:

Source	SS	df	MS	F
FactorA	29.27521	1	29.27521	5.532831996
FactorB	454.69923	3	151.56641	28.64510563
FactorA & B	27.34379	3	9.114596667	1.72260189
Error	169.3	32	5.29118	
Total	680.63599	39		

(b) $F = 1.723 < 2.90 = F_{\{3,32\}}$, we fail to reject H_{AB}

(c) $F = 5.53 > 4.15 = F_{\{1,32\}}$, we reject H_A

(d) $F = 28.645 > 2.90 = F_{\{3,32\}}$, we reject H_B

3. A sports scientist suspects that the ranking of a tennis player is affected by the injury level and height of the player. He selects 3 levels of injury and 5 height ranges. He conducts an experiment and obtains the ANOVA table as follows.

Source	SS	df	MS	F
Injury level			2228.6333	
Height				
Interaction	7128.6667			
Error			942.7667	
Total	27900.1667	29		

The experiment can be modeled as

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}, \quad \begin{cases} i = 1, 2, ..., a \\ j = 1, 2, ..., b \\ k = 1, 2, ..., c \end{cases}$$

where α_i denotes effect of the ith injury level, β_j is the effect of the jth height range. $\epsilon_{ijk} \overset{i.i.d.}{\sim} N(0, \sigma^2)$.

(a) Specify the values for a, b, and c.

(b) Finish the AVOVA table.

(c) Conduct all the hypothesis tests corresponding to different effects and draw your conclusion.

(a) $a = 3$, $b = 5$, $abc - 1 = 29 \Rightarrow c = 2$.

(b) The AVOVA table is

Source	SS	df	MS	F
Injury level	4457.2667	2	2228.6333	2.3639
Height	2172.7329	4	543.1832	0.5762
Interaction	7128.6667	8	891.0833	0.9452
Error	14141.5005	15	942.7667	
Total	27900.1667	29		

(c) To test the effect of injury level, we have

$$H_0^A : \alpha_1 = \alpha_2 = \alpha_3 = 0, \quad H_1^A : \text{ Not all } \alpha_i, i = 1, 2, 3 \text{ are } 0.$$

Since $2.3639 < F_{0.05}(2, 15) = 3.6823$, we fail to reject H_0^A.

To test the effect of height range, we have

$$H_0^B : \beta_1 = \ldots = \beta_5 = 0, \quad H_1^B : \text{ Not all } \beta_i, i = 1, 2, \ldots, 5 \text{ are } 0.$$

Since $0.5762 < F_{0.05}(4, 15) = 3.0556$, we fail to reject H_0^B.

To test the effect of interaction, we have

$$H_0^{AB} : (\alpha\beta)_{ij} = 0, \quad H_1^{AB} : \text{ Not all } (\alpha\beta)_{ij} \text{ are } 0.$$

Since $0.9452 < F_{0.05}(8, 15) = 2.6408$, we fail to reject H_0^{AB}.